

The Role of Peers' Executive Function and Classroom Quality in Preschoolers' School Readiness

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ABSTRACT

The present study examined associations between peers' EF and gains in individual children's EF, language, mathematics, and literacy skills across the preschool year, and whether these associations are moderated by classroom quality. Participants included three cohorts of children ($n = 556$, 52.5% male, $M_{age} = 57.5$), their parents, and teachers from a larger evaluation of a state-funded prekindergarten program. Multilevel models indicated that peers' EF in the fall of preschool was a significant predictor of individual children's EF ($\beta = 0.14$, $p = .001$), mathematics ($\beta = 0.06$, $p = .049$) and literacy ($\beta = 0.09$, $p = .027$) in the spring of preschool, but was not related to language ($\beta = 0.03$, $p = .342$), while controlling for child and classroom level covariates, fall EF, and relevant pre-academic and language skills. Results from moderation analyses indicated a significant interaction effect between classroom quality and peers' EF in predicting children's literacy ($\beta = 0.07$, $p = .040$), suggesting that when children are in low-quality classrooms, having peers with higher EF was more strongly related to gains in literacy skills. No other significant interaction effects were found between classroom quality and peers' EF in predicting individual children's EF, language, and mathematics. Implications for future research on peer effects and practice and policy for preschool education are discussed.

Executive function (EF), a set of higher-order cognitive skills that help children regulate their behaviors, attention, and emotions, is an important predictor of school readiness (Blair, 2002; Diamond, 2013; McClelland et al., 2013). Children practice their EF skills with adults and peers in preschool settings (Weiland et al., 2013a). Although the evidence between the associations of classroom quality and children's EF is mixed (see Guerrero-Rosada et al., 2021), a high-quality preschool environment, as defined by positive teacher-child interactions and the presence of stimulating and enriching learning opportunities, may support children's EF (Perlman et al., 2016; Weiland et al. 2013a). When teachers provide high-quality instructions and support self-regulation, children benefit from learning opportunities and gain school readiness skills, including pre-academic and language skills (Burchinal et al., 2010). Even though classroom quality and teacher-child interactions have been examined as predictors of EF and school readiness skills, less is known about the potential role of peers in the development of children's EF and pre-academic skills during preschool (Skibbe et al., 2012).

Peer interactions are essential for children's development (Bronfenbrenner & Morris, 2007; Vygotsky, 1987). The influence of peers' skills on individual children's development may be especially prominent in preschool where children are encouraged to engage in social, interactive play – play and interact with peers rather than solitary or parallel play – playing apart or beside, but with not with other children (Coplan et al., 2015; Eggum-Wilkens et al., 2014). In fact, peer interactions tend to occur naturally within the preschool classroom and learning occurs during these peer interactions (Bulotsky-Shearer et al., 2014; Hanish et al., 2005). Prior studies have shown that positive peer interactions during play such as being cooperative and resolving conflicts are related to academic skills (Fantuzzo et al., 2004). In preschool settings where group participation likely influences classroom activities (e.g., waiting until the group becomes quiet to start an activity), peers' EF skills may be particularly important for individual children's development of EF, as well as pre-academic and language skills. Despite the emerging evidence of peer effects on children's academic skills (Choi et al., 2018; Montroy

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et al., 2016; Wei et al., 2021), studies to date focus only on literacy and language skills, not on mathematics, and findings from this work have been inconsistent. Understanding the early precursors of mathematics skills is important due to its strong predictive role in later academic skills (Claessens & Engel, 2013; Duncan et al., 2007). In addition to direct relations between peers' skills and children's development, peers' skills may also interact with classroom quality. Peer interactions are essential contributors to the organization of classroom activities (Bulotsky-Shearer et al., 2014). Yet, to our knowledge, no studies have examined whether classroom quality interacts with classroom composition of EF (i.e., peers' EF) to predict gains in individual children's EF, language, and pre-academic skills. Therefore, in this study, we contribute to this emerging area of research by examining 1) whether peers' EFs are related to gains in individual children's EF, language, and pre-academic (i.e., mathematics and literacy) skills across the preschool year and 2) whether classroom quality moderates the associations between peers' EF and individual children's EF, language, and pre-academic skills.

The Role of EF in the Preschool Classroom

EF is broadly defined as a set of cognitive processes that are central to the development of young children's cognitive and social competence and the emerging self-regulation of behavior (Blair, 2002). Children's EF shows a substantial development in the preschool years (Welsh et al., 2010). Researchers argue that EF consists of three distinct, yet related factors: inhibitory control, working memory, and cognitive flexibility (Best & Miller, 2010). Inhibitory control refers to one's ability to suppress attention and behavior and respond to a stimulus (Zelazo et al., 2016). Working memory involves both the retention and manipulation of information (Zelazo et al., 2016). Cognitive flexibility, also referred to as attention shifting, includes the ability to direct one's attention as required to a particular stimulus (Morrison et al., 2010).

Theoretical and empirical literature highlights the importance of EF for children's learning and various short- and long-term outcomes. It has been posited that well-developed EF skills in early childhood help children to take advantage of learning opportunities in a classroom context (Blair & Raver, 2015). Consequently, a growing number of empirical studies has shown that EF skills predict early academic achievement and long-term school success (Blair & Razza, 2007; Duncan et al., 2007; McClelland et al., 2007). Specifically, EF is associated with mathematics (Allan & Lonigan, 2014; Schmitt et al., 2017), literacy (Fuhs et al., 2014), and language (Weiland et al., 2013b) in preschool, with stronger links documented between EF and mathematics (Allan & Lonigan, 2014; Schmitt et al., 2014).

The Role of Peers' EF in the Development of EF, Language, and Pre-academic skills

Importance of Peers on Children's Development

Peer relationships play an important role in children's development (Asher et al., 1990; Hartup, 1996). Developmental and learning theories have emphasized that children's development occurs through socially mediated processes. Social learning theory, for instance, suggests that children learn by observing people around them behaving in different ways, such that peers' skills and behaviors could directly impact children's cognitive and behavioral learning during peer play and interaction (Bandura, 1997; Choi et al., 2018). Social modeling can be a powerful tool; if children have multiple opportunities to witness positive reinforcement from a particular type of behavior, they are more likely to adopt that behavior themselves and less likely to perform negatively reinforced behaviors (Bandura, 1997). Further, sociocultural theory views human development as a socially mediated process and suggests the influence of direct peer effects on learning when children interact with and are challenged by more capable peers (Berk & Winsler, 1995; Choi et al., 2018; Vygotsky, 1987). As children transition from toddlers

to preschoolers, they start to transition from solitary and parallel play to social and interactive play (Coplan et al., 2015). It is also during ages 3 to 5 that children's EF skills show substantial improvement (Howard et al., 2015) and make social and interactive play an avenue for learning and development (Bulotsky-Shearer et al., 2014). Therefore, peer effects literature suggests that children's development in a given developmental domain, such as academic skills, will be affected by the skills within the same domain of their peers with whom a child has daily contact (Choi et al., 2018). In addition to within-domain effects, emerging research with preschool children has shown cross-domain peer effects. For instance, children in classrooms where peers have higher levels of EF demonstrate greater gains in EF and literacy skills in the preschool (Montroy et al., 2016) as well as in the first grade (Skibbe et al., 2012). Thus, children's development of EF may be particularly important for both their own and their peers' acquisition of EF as well as pre-academic and language skills.

Relations between Peers' EF and Children's EF, Language, and Pre-academic skills

In a classroom setting, interacting with peers with high levels of EF provides children with examples of well-regulated behavior, more opportunities to practice self-regulation skills, and the motivation to regulate themselves (Finch et al., 2019). For instance, one study that utilized a low-income sample found that children in classrooms where peers had higher EF skills demonstrated greater gains in teacher-reported EF skills at the end of the school year (Choi et al., 2018). Similarly, another study found that the effect of peers' EF was greater for children with lower levels of EF compared with children with higher levels of EF (Montroy et al., 2016). Additionally, higher average levels of classmates' EF, as measured by students' reaction times on direct EF assessments, were linked to increased growth among individual students' reaction times on EF tasks in elementary school (Finch et al., 2019). Despite these significant findings, there is also evidence for non-significant associations between peers' EF and individual children's EF. For instance, an intervention study aimed to improve children's EF skills for the entire classroom found no significant links between peers' EF and individual children's EF (Weiland & Yoshikawa, 2014).

Language is an important component of school readiness and is foundational for reading achievement (Duncan et al., 2007). Although not many studies have specifically examined the cross-domain associations between peers' EF and language skills, theoretical and empirical perspectives (Justice et al., 2011; Mashburn et al., 2009) indicate the potential connection between peers' EF and children's language development. It is argued that children's early language acquisition can be positively stimulated by their peers' EF skills. In preschool settings, children are often engaged in collaborative and group play, which necessitates well-developed EF skills. Children with well-developed EF skills engage in peer-to-peer conversations with classmates and provide linguistic inputs to each other in play groups (Shaheen, 2014), which may directly promote children's language development in preschool classrooms. In addition, peers with well-developed EF skills may use frequent self-talk during play (Fahy, 2014), which also may model the frequent use of language for peers. Although prior evidence suggests a potential link between peers' skills and language skills as measured by expressive and receptive vocabulary (Mashburn et al., 2009), several recent studies examining cross-domain links have documented no associations between peers' EF and children's language as measured by their receptive vocabulary skills (Choi et al., 2018; Weiland & Yoshikawa, 2014), thus the relations between the two remain unclear.

Previous research has shown comparable relations between peers' EF and children's pre-academic skills, such as mathematics and literacy. For example, in one study, classroom peers' EF was found to be associated with individual growth in early letter word decoding, but not letter name knowledge gains during preschool (Montroy et al., 2016). Additionally, peer effects continued to be observed as children transition

into elementary school. For instance, a large-scale longitudinal study showed that peer levels of EF in preschool could predict a child's literacy gains in first grade (Skibbe et al., 2012).

Despite the growing prior work of peers' EF on children's EF, literacy, and language skills, there are no studies examining the extent to which these associations extend to mathematics. EF is a strong predictor of mathematics (Blair & Razza, 2007; Bull et al., 2008; Cragg & Gilmore, 2014; McKinnon & Blair, 2019) likely because learning mathematics involves mastering steps toward achieving a goal, such as coming to the solution of a problem or a puzzle (Best et al., 2011; Clements et al., 2016; McClelland et al., 2014; Schmitt et al., 2017). In a classroom context, observing peers with high EF skills, such as engaging in planning related activities (e.g., block building), may contribute to early mathematics skills. Given the critical role of EF in the development of mathematics proficiency (Cragg & Gilmore, 2014), it is important to understand the role of peers' EF on children's emerging mathematics skills. Based on the observed associations between peers' EF and literacy, we hypothesize that in a similar manner, peers' EF may play a role in children's individual mathematics learning. Understanding the unique associations between peers' EF and emerging mathematics skills may add to the current body of knowledge related to the development of EF and mathematics skills. These associations may also inform targeted interventions aiming to boost children's early mathematic skills.

The Role of Classroom Quality and EF, Language, Pre-academic Skills

Peers' influence on children's development cannot be considered outside of the context in which it occurs. In a classroom setting, children are consistently expected to modify their behaviors according to the rules and routines set by teachers as well as peers (Kern & Clemens, 2007). The quality of the classroom environment, defined by high-quality teacher-child interactions, has been studied as an important predictor of children's development of various skills (Connor et al., 2009; Hamre & Pianta, 2007; Vandenbroucke et al., 2018). There are three specific domains of classroom quality that have been identified in past literature as being particularly important – emotional support, classroom organization, and instructional support (La Paro et al., 2004), although recent evidence with regard to their links to children's school readiness skills is mixed (Perlman et al., 2016). For instance, some studies documented evidence for the instructional support domain and children's executive function, mathematics, literacy, and language (Hamre et al., 2014), and some provided very small or null findings (Burchinal, 2018; Guerrero-Rosada et al., 2021).

A high-quality classroom environment may be indicative of a classroom that is more effectively meeting the needs of diverse groups of children (e.g., scaffolding, individual time), whereas a lower-quality classroom environment may be indicative of a classroom where more time is spent on classroom management and organization instead of instructional time (Marzano et al., 2005; Stronge & Hindman, 2003). Peer effects on individual children's outcomes could be amplified by the quality of the classroom environment. Previous research has shown that student composition characteristics such as gender (Lavy, 2015), varying levels of achievement (Aucejo et al., 2018; Connor et al., 2014), socio-economic status (Konstantopoulos, 2009), and age (Ansari & Pianta, 2018) have implications on perceived classroom quality. Despite other aspects of classroom composition being studied in relation to classroom quality, it is unclear whether the association between classroom composition of EF skills and children's individual outcomes will be attenuated or amplified for classrooms with higher or lower quality.

There is scant research concerning the moderation between classroom quality and peer effects. However, classroom quality is important to study in the peer effect context due to its potential to impact children's various skill development as well as child-to-child transmission of skills (Bulotsky-Shearer et al., 2014; Henry & Rickman, 2007). Given that peer interactions are key to the organization of structured or

unstructured classroom activities in early childhood classrooms, the extent to which teachers organize the classroom environments may influence the effect of peers' skills on individual children's EF, pre-academic, and language skills.

There could be several different ways peers and classroom quality may interact and affect children's outcomes. In classrooms where there is a low-quality environment, peer EF may be more related to children's outcomes than where there is a high-quality environment. For instance, in a low-quality classroom environment, the teacher may spend more time on behavior management and less on instructional activities, and children could rely on their peers' skills to engage in learning activities. Conversely, in a high-quality classroom environment, teachers may successfully promote self-regulation and use more time on instructional activities, and peers' skills may not be as important for children to benefit from learning opportunities. Alternatively, peer EF may be more related to children's outcomes when there is a high-quality environment. For example, peer relations may be more conflictual in a low-quality classroom environment and children may not benefit as much from their interactions with peers and cannot engage in learning activities. However, in a high-quality classroom environment, instructional activities may necessitate peer interactions that require regulatory skills and children may benefit from peers' skills. Despite the potential interplay between peers' EF and classroom quality (Yeomans-Maldonado et al., 2019), its role in individual children's EF, language, and pre-academic skills development over the preschool year has remained unexplored. Exploring the extent to which classroom quality influences the effects of peers' skills on individual children's development could provide findings for new avenues of research and policy implications targeting preschool education practices.

Present Study

The goals of the present study are to expand upon the emerging previous work (e.g., Choi et al., 2018; Montroy et al., 2016) that examined the associations between peers' EF and individual children's EF, language, and pre-academic skills (e.g., mathematics and literacy) in preschool. Our first aim was to examine the associations between peers' EF and individual children's EF, language, mathematics, and literacy skills. Based on previous findings, it was expected that children in classrooms where peers have a higher average level of EF would demonstrate greater gains in individual EF, language, mathematics, and literacy skills. Our second aim was to investigate whether the effects of peers' EF on children's development of individual EF, language, mathematics, and literacy skills were moderated by classroom quality. Because there are no studies that have explored the combined effects of peers' EF and classroom quality on children's development, we did not have an *a priori* hypothesis. Instead, we acknowledge that multiple hypotheses could be supported that are theoretically driven. For example, it may be that associations between peer EF and gains in the target outcomes are stronger for those children who are in high-quality classrooms. Or it could be that these associations are weaker for children in high-quality classrooms. Similarly, it may be that associations become stronger or weaker in low-quality classrooms.

Method

Participants

The participants ($n = 684$) for this study came from a larger study evaluating a state-funded prekindergarten program in a midwestern region of the United States. The study included children who participated in the state-funded prekindergarten program ($n = 456$) and children who attended other community-based preschool programs ($n = 228$). The sample included three cohorts of children, their parents, and their teachers who were followed during the preschool year. Cohort 1 consisted of children who attended preschool from 2015 to 2016 ($n =$

371), Cohort 2 from 2016 to 2017 ($n = 187$), and Cohort 3 from 2018 to 2019 ($n = 126$). The study only included children from classrooms that had three or more participating children, and on average, there were seven participating children ($SD = 3$, min-max = 3–20) from each classroom, which is in line with previous peer effects studies that typically examined four to seven children per classroom (Choi et al., 2018; Justice et al., 2011). The total study sample consisted of 556 children (52.5% male, $M_{age} = 57.5$ months) from two groups, state-funded Pre-K ($n = 387$) or comparison ($n = 190$). All children met an income eligibility requirement for program participation (at or below 127% of the federal poverty level). Among the sample, 44% were African American, 32% of the children were White, 13% were Hispanic, and 11% were multi-racial. The average monthly income of parents was \$1532 ($SD = \892).

Procedures

The data collection procedures for all child and classroom data were the same across the cohorts. Research assistants administered a battery of developmental assessments to each child at their school, in both the fall and spring of preschool, which lasted approximately 60–90 min. Assessments took place in children's preschools in a quiet space designated by the preschool directors or teachers. Research assistants were trained by the lead project staff to administer child assessments and became reliable on administration. Research assistants conducting classroom observations using the Classroom Assessment Scoring System, Pre-K (CLASS Pre-K; Pianta et al., 2008) participated in two days of training with Teachstone and completed a test to become reliable observers. Following training and reliability testing, four, 15-min observations were conducted in each classroom, assessing the quality of the classroom environment during the winter of preschool. Parents provided demographic information (age, sex, income, race) through questionnaires.

Measures

Executive Function

The Head-Toes-Knees-Shoulders task (HTKS; McClelland et al., 2014) was used to assess the children's EF skills. The HTKS is a direct behavioral measure that taps into all three components of EF (inhibitory control, working memory, and cognitive flexibility), and is typically used with children between the ages of three and seven. The HTKS is presented as a game where the children are asked paired combinations of four rules (i.e., "touch your head," "touch your toes," "touch your knees," and "touch your shoulders"). In the practice round, children are asked to respond by following the directions naturally, and then they are asked to respond in the opposite way (e.g., children are asked to touch their toes when the research assistant says "touch your head"). The testing portion consists of 30 items (three sections of 10), that become increasingly complex as the child progresses through the assessment. Each correct response is worth two points as each item is scored as 0 (incorrect), 1 (self-correct), or 2 (correct). Consequently, the possible scores range from 0 to 60. The total score of the test is the sum of all the correct items. To progress to the second section, a child must receive a score of at least 4 on the first section, and likewise, to progress to the third section, a child must receive a score of at least 4 on the second section. The HTKS takes about 5–10 min to complete. The interrater reliability, scoring agreement, and test-retest reliability is high and shows strong predictive validity (McClelland et al., 2014), and the task showed strong reliability (i.e., internal consistency) in the fall (Cronbach's $\alpha = 0.96$) and spring (Cronbach's $\alpha = 0.97$) of preschool.

Peers' Executive Function

Following prior work (Justice et al., 2011; Montroy et al., 2016; Skibbe et al., 2012), peers' EF was calculated by taking the skill levels of participating children in a classroom as a whole. Specifically, the

average fall level of EF by classroom is calculated to represent peers' EF.

Language

Children's knowledge of spoken words and receptive vocabulary was assessed using the Peabody Picture Vocabulary Test, fourth edition (PPVT-4; Dunn & Dunn, 2007). This test draws on the cognitive processes involved in matching verbal cues to pictures, using no memory or recall components. The PPVT-4 is comprised of 228 items grouped into 19 sets of 12 items per set. The PPVT-4 has strong internal consistency ($\alpha = 0.94$) and takes about 10–15 min to administer (Dunn & Dunn, 2007). The research assistant presents a series of four images from the test booklet to the child. The research assistant pronounces a word (receptive component) and then the child points to which of the four images shown matches that word. The child continues from section to section until they respond incorrectly eight times in one section. To adjust for differing ages of the children, the research assistants use the basal rule and start the task at various sections depending on the age of the child. A higher score on the task indicates better receptive vocabulary. Raw scores were converted to age-based standard scores and standard scores were used in analyses. The assessment demonstrated strong reliability for the current sample (Cronbach's α for internal consistency for the fall = 0.97 and Cronbach's α for internal consistency = 0.97 for the spring).

Early Mathematics Skills

Children's mathematics skills were assessed using the Applied Problems subtest of the Woodcock-Johnson Test of Achievement-IV (WJ-IV; Schrank et al., 2014) and the Preschool Early Numeracy Skills – Brief Version (PENS; Purpura et al., 2015). The Applied Problems subtest of the WJ-IV is a task that assesses early mathematical operations (e.g., counting, addition, subtraction) using practical problems. The Applied Problems subtest contains 63 items and takes about 5–15 min to administer. During this task, children were asked to respond to questions until they provided five consecutive incorrect responses at the end of a set. Children received one point for each correct response. The assessment demonstrated strong reliability for this age group (Villarreal, 2015).

The PENS measures children's early numeracy skills including set comparisons, numeral comparisons, one-to-one correspondences, number order, numeral identification, ordinality, and number combinations, with 24 items ordered by difficulty, with the easiest items asked first. The task takes 5 min to administer. During the task, children are shown a picture and asked to point to the correct response or provide a verbal response until the child provided three consecutive incorrect responses. Children received one point for each correct response. The assessment demonstrated strong reliability for the current sample (Cronbach's α for internal consistency = 0.89 in the fall and Cronbach's α for internal consistency = 0.89 in the spring). Given the strong correlation between the two measures in the fall ($r = 0.70$) and spring ($r = 0.67$), a math composite score was created by taking the average of z-scores of the WJ-AP and the PENS and a composite score was used in the analyses.

Early Literacy Skills

Children's literacy skills were assessed using the Letter-Word Identification subtest of the Woodcock-Johnson Tests of Achievement-IV (WJ-IV; Schrank et al., 2014). The Letter-Word Identification subtest of the WJ-IV measures early literacy skills (e.g., letter identification, word reading, and reading aloud) using expressive and receptive items. The Letter-Word Identification subtest contains 76 items and takes about 5–15 min to administer. Children received one point for each correct response. Research assistants stopped administering the assessment if a child incorrectly responded to six consecutive items at the end of a set. The assessment demonstrated strong reliability for this age group (Villarreal, 2015).

Classroom Quality

The quality of the classroom was assessed using the Classroom Assessment Scoring System (CLASS), which is an observational tool that is used to assess global classroom processes. The classroom processes are assessed in three domains: (1) Emotional Support, (2) Classroom Organization, and (3) Instructional Support. Each domain contains a set of dimensions with observable behavioral markers. The Emotional Support domain includes the dimensions of positive climate, negative climate, regard for student perspective, and teacher sensitivity. The Classroom Organization domain includes the dimensions of behavior management, instructional learning formats, and productivity. The Instructional Support domain includes the dimensions of concept development, language modeling, and quality of feedback.

The CLASS involves a sequence of four 15-min observations during which the certified CLASS observer takes notes on the quality and frequency of teacher-child interactions. After each observation there is a 10-min coding session during which the certified CLASS observer consults the CLASS manual to assign scores to the ten dimensions on a 7-point Likert scale (1–2 = low quality; 3–5 = medium quality; 6–7 = high quality). To compute the CLASS scores that are used in analyses, each dimension is averaged across four intervals, and then the average dimensions for each domain are averaged together to represent the final score for each of the three CLASS domains. The average score across the three CLASS domains is used in the analyses due to the high correlation between the domains (ranging between 0.52 and 0.76). The CLASS has been demonstrated to be reliable and valid in previous work (Cloney et al., 2017). The Emotional Support, Classroom Organization, and Instructional Support domains of the CLASS yielded high internal consistency in the current sample (Emotional Support = 0.94; Classroom Organization = 0.92, and Instructional Support = 0.92).

Covariates

Children's age in months, sex (0 = male; 1 = female), family income, cohort status, group status (0 = comparison, 1 = state Pre-K), and fall EF, and relevant pre-academic and language scores were used as control variables in models. Peer covariates included classroom age and classroom income, and classroom quality was also included as a covariate.

Analytic Strategy

Multilevel Modeling

In the current study, peer EF and classroom quality are both classroom-level predictors, and children's data within classrooms are not independent. Therefore, multilevel modeling was used to correctly identify the variation that exists at the classroom- and child-level in the dataset using Stata 16.0 (StataCorp., 2019). We used a series of two-level models because children were nested within classrooms. This approach allowed us to correctly specify the effects of both child-level (e.g., children's own baseline skills) and classroom-level (e.g., peer EF) vari-

As a second step, main effect models were run to examine the effects of peer EF in the fall on children's outcomes in the spring. The grand mean centered average peer levels of EF were included to the second level (i.e., classroom level) to estimate the extent to which gains in individual children's skills was related to peers' EF skills. Specifically, four main effects models were run separately for each outcome (i.e., EF, language, early mathematics, and early literacy). In these models, we controlled for child-level data (i.e., baseline of the outcomes variables), child-level EF, and a host of child, peer, and family covariates in the fall, which allowed us to look at any residualized change in the outcome variables in the spring. Including child-level fall EF in all the models in addition to the specific baseline skill that is examined was important due to strong associations between EF and language and pre-academic skills in this age group (Blair & Razza, 2007; Purpura et al., 2017; Slot & von Suchodoletz, 2018). For instance, we included fall EF in the model when predicting spring mathematics in addition to the baseline mathematics (i.e., fall mathematics). Therefore, the models account for children's beginning-year levels of EF, and the associations observed at the end of the year are due to changes in the outcomes beyond what would be due to the associations between initial levels of EF and outcome variables. In all models, we also included a robust set of covariates guided by previous research that represents the characteristics of children and their families, peers, and classrooms. Characteristics of children and families (i.e., age at fall assessment, sex, family income) were included as Level 1 variables. Classroom level covariates (classroom average age at fall assessment, average family income, cohort assignment, and group status) were included as Level 2 variables. Finally, as suggested in previous literature, we included the quality of classroom as a classroom-level covariate in all models to ensure that our estimates of the peer effects were not confounded by classroom-level quality (Choi et al., 2018; Justice et al., 2011; Montroy et al., 2016; Weiland & Yoshikawa, 2014). All continuous variables in the equations were grand mean centered while examining the influence of Level 2 variables controlling for Level 1 variables. Since we were interested in whether classroom-level EF is associated with children's outcomes in addition to the effect at the individual level, grand mean centering is recommended by Enders and Tofghi (2007), Lüdtke et al. (2009), and prior research (Choi et al., 2018; Montroy et al., 2016). It is important to note that centering in multilevel models gives a more meaningful interpretation of the coefficients and variance components (Raudenbush & Bryk, 2002). In addition, all continuous variables were z-scored to receive standardized coefficients (e.g., effect sizes). The final main effect model for EF was as follows:

Level 1: Y_{ij}

$$= \beta_{0j} + \beta_{1j} * (\text{fall EF}_{ij}) + \beta_{2j} * (\text{age}_{ij}) + \beta_{3j} * (\text{sex}_{ij}) + \beta_{4j} * (\text{income}_{ij}) + r_{ij}$$

$$\text{Level 2: } \beta_{0j} = \gamma_{00} + \gamma_{01} * (\text{peer age}_j) + \gamma_{02} * (\text{peer income}_j) + \gamma_{03} * (\text{peer EF}_j) + \gamma_{04} * (\text{CLASS}_j) + \gamma_{05} * (\text{cohort}_j) + \gamma_{06} * (\text{group}_j) + u_{0j}$$

ables on the outcomes of interest. First, unconditional models were used to calculate the intraclass correlations (ICCs) for each outcome to examine the extent to which children's skills differed between classrooms. ICCs were 0.13 for EF, 0.15 for language, 0.15 for mathematics, and 0.29 for literacy in the spring, indicating that children's skills differed between classrooms and about 13 to 29% of the variance in children's skills is at the classroom level. Additionally, ICC₂ was calculated for the peer EF variable to provide an estimate of the reliability of this aggregate variable. ICC₂ was 0.74, indicating a sufficient degree of reliability for the peer EF variable (Lüdtke et al., 2009).

where a spring EF score (Y) for a child (i) in classroom (j) is defined as a function of the intercept, representing the classroom mean score in the spring, individual EF score in the fall, child and family level covariates (age, sex, income, cohort, group), peer covariates (peer age, peer income), peer EF, classroom quality, and the error term. As a final step, moderation effects were run to examine whether peer effects of EF on children's outcomes varied by the quality of classroom experiences. Specifically, we added an interaction term at Level 2 between peers' EF and classroom quality to each of our models.

Missing Data

Missing data was handled by using multiple imputation (MI) in Stata 16.0 (StataCorp., 2019). Specifically, EF in the spring was missing for 12.3% of the sample, mathematics in the spring was missing for 10.5% of the sample, literacy was missing for 9.8% of the sample, and language was missing for 12.1% of the sample. All other study variables were not missing or were missing for less than 5% of the sample. Prior to estimating models, MI with chained equations was used to impute 25 data sets and all variables that were included in the final model were included in the imputation models. The results, parameter estimates, and standard errors were combined across the imputed datasets.

Results

Descriptive Statistics

Descriptive statistics can be found in Table 1 and correlations can be found in Table 2. Peer EF in the fall was significantly correlated with individual EF, language, mathematics, literacy in the fall as well as in the spring. However, peer EF was not significantly associated with classroom quality. Individual EF, language, mathematics, and literacy were all significantly correlated with each other in the fall and in the spring. Classroom quality was only significantly correlated with group status and peer age in the fall and not significantly correlated with any other variables.

The effects of Peer EF on Gains in Individual Children's EF, Language, and Pre-academic Skills

A multilevel model framework was used to explore whether there was an association between peers' EF and children's gains in EF, language, and pre-academic skills during preschool (see Table 3). As

Table 1
Descriptive Statistics for Study Variables

Variables	n	Mean (SD)/%	Range
Covariates			
Fall age in months	577	57.56 (3.73)	44.84–70.62
Spring age in months	542	62.50 (3.69)	49.48–74.17
Sex (1 = female)			
Male	302	52.50%	
Female	275	47.50%	
Income	567	1532.59 (892.56)	0–6257.77
Fall peer age	577	57.56 (2.04)	51.32–63.33
Peer income	579	1531.23 (436.11)	596.20–3036.46
Cohort (1 = expansion)			
Cohort 1 (pilot)	327	56.48%	
Cohort 2 (pilot)	159	27.46%	
Cohort 3 (expansion)	93	16.06%	
Group (1 = state Pre-K)			
Comparison	190	32.82%	
State Pre-K	387	67.18%	
Peer EF	577	11.31 (8.04)	0–36.5
Children's EF			
Fall EF	556	11.26 (15.05)	0–59
Spring EF	504	18.36 (18.50)	0–60
Children's language			
Fall language	542	93.76 (14.96)	27–130
Spring language	506	98.08 (15.76)	47–140
Children's pre-academic skills			
Fall mathematics	556	403.35 (21.11)	272–448
Fall literacy	562	334.72 (25.95)	249–465
Spring mathematics	514	412.33 (16.58)	324–453
Spring literacy	521	345.61 (26.89)	272–467
Classroom characteristics			
CLASS	571	4.03 (0.87)	1.96–5.88
Emotional Support	571	5.24 (1.00)	2.81–6.88
Classroom Organization	571	4.62 (1.14)	1.42–6.75
Instructional Support	571	2.22 (0.89)	1–5.17

Note. EF = Executive function, CLASS = Classroom Assessment Scoring System.

Table 2
Correlations among Study Variables

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.	17.
1. Fall age	–																
2. Sex	0.01	–															
3. Income	–0.01	–0.03	–														
4. Cohort	–0.09*	–0.07	0.12**	–													
5. Group	–0.03	0.03	–0.15***	–0.12**	–												
6. Fall peer age	0.55***	0.03	–0.01	–0.15**	–0.30***	–											
7. Fall peer income	–0.01	–0.04	0.48***	0.25***	–0.08	0.00	–0.03	–									
8. Fall EF	0.20***	0.10*	0.00	–0.08	0.14**	0.19***	–0.06	0.53***	–								
9. Fall peer EF	0.15***	0.02	–0.02	–0.14**	0.00	0.19***	0.10*	0.42***	0.31***	–							
10. Fall mathematics	0.27***	0.08	0.03	–0.05	0.02	0.19***	–0.06	0.33***	0.24***	0.43***	–						
11. Fall literacy	0.19***	0.05	0.06	–0.11**	0.02	0.19***	0.10*	0.33***	0.34***	0.19***	0.43***	–					
12. Fall language	0.10*	0.07	0.00	–0.07	0.02	0.10*	–0.06	0.44***	0.18***	0.28***	0.54***	0.11*	–				
13. Spring age	0.97***	0.01	–0.03	–0.07	0.01	0.50***	–0.01	0.22***	0.34***	0.45***	0.37***	0.46***	0.23***	–			
14. Spring EF	0.21***	0.05	0.02	–0.05	–0.03	0.14**	–0.03	0.64***	0.43***	0.68***	0.49***	0.52***	0.26***	0.50***	–		
15. Spring mathematics	0.25***	0.06	0.04	–0.09*	–0.03	0.19***	0.03	0.47***	0.37***	0.68***	0.54***	0.54***	0.34***	0.33***	0.49***	–	
16. Spring literacy	0.21***	0.09*	0.03	–0.15***	–0.13**	0.14**	0.09*	0.31***	0.27***	0.40***	0.75***	0.34***	0.22***	0.33***	0.54***	0.34***	–
17. Spring language	0.13**	0.10*	–0.02	–0.21***	0.04	0.14**	–0.09	0.43***	0.37***	0.53***	0.40***	0.77***	0.13**	0.48***	0.54***	0.34***	–
18. CLASS	0.07	0.02	–0.02	0.01	0.13**	0.14**	–0.05	–0.00	–0.00	–0.00	0.08	0.05	0.03	0.01	–0.03	0.03	0.03

Note. * $p < .05$, ** $p < .01$, *** $p < .001$.

Table 3
Model estimates for main and interaction effects of peer EF and classroom quality on children's executive function, math, literacy, and language ($n = 577$).

Variables	EF				Mathematics				Literacy				Language			
	Main effect model		Interaction effect model		Main effect model		Interaction effect model		Main effect model		Interaction effect model		Main effect model		Interaction effect model	
	β	SE	β	SE	β	SE	β	SE	β	SE	β	SE	β	SE	β	SE
Intercept	−0.08	0.11	−0.08	0.11	0.08	0.08	0.09	0.08	0.14	0.10	0.14	0.10	0.23**	0.08	0.23**	0.09
Peer (fall) EF	0.14**	0.04	0.13**	0.04	0.06*	0.03	0.06*	0.03	0.09*	0.04	0.09*	0.04	0.03	0.03	0.03	0.03
Child (fall) EF	0.54***	0.04	0.54***	0.04	0.09**	0.03	0.09**	0.03	−0.02	0.03	−0.02	0.03	0.07*	0.03	0.07*	0.03
Mathematics					0.74***	0.03	0.74***	0.03								
Literacy									0.75***	0.03	0.75***	0.03				
Language													0.72***	0.03	0.72***	0.03
Covariates																
Age	0.08*	0.04	0.08*	0.04	0.01	0.03	0.01	0.03	0.04	0.03	0.04	0.03	0.00	0.03	0.00	0.03
Income	0.05	0.04	0.05	0.04	0.07**	0.03	0.07**	0.03	0.01	0.03	0.01	0.03	0.02	0.03	0.02	0.03
Sex (female)	−0.03	0.05	−0.04	0.07	−0.01	0.04	−0.01	0.04	0.06	0.05	0.06	0.05	0.06	0.06	0.06	0.06
Cohort (expansion)	0.03	0.05	0.03	0.05	−0.03	0.03	−0.03	0.03	−0.14**	0.05	−0.14**	0.04	−0.21***	0.04	−0.21***	0.04
Group (state Pre-K)	0.05	0.08	0.05	0.08	−0.04	0.05	−0.04	0.05	0.09	0.07	0.09	0.07	0.09	0.06	0.09	0.06
Peer age	−0.03	0.04	−0.03	0.04	0.01	0.03	0.01	0.03	−0.05	0.04	−0.06	0.04	0.03	0.03	0.03	0.03
Peer income	−0.02	0.04	−0.02	0.04	0.00	0.03	0.00	0.03	0.04	0.04	0.04	0.04	0.01	0.03	0.01	0.03
CLASS	0.01	0.11	0.01	0.11	−0.04	0.03	−0.04	0.03	−0.02	0.03	−0.03	0.03	−0.01	0.03	−0.01	0.03
Peer EF x CLASS			−0.04	0.03			−0.02	0.02			−0.07*	0.03			0.02	0.03
Variance components (random effect)		Estimate (SE)				Estimate (SE)				Estimate (SE)				Estimate (SE)		
Intercept (τ_{00})		0.00 (0.00)				0.11 (0.04)				0.20 (0.04)				0.00 (0.00)		
Residual (σ^2)		0.77 (0.02)				0.49 (0.02)				0.58 (0.02)				0.61 (0.02)		

Note. EF = Executive function, CLASS = Classroom Assessment Scoring System.

* $p < .05$, ** $p < .01$, *** $p < .001$.

expected, the analyses indicated that peers' EF in the fall was positively related to individual children's EF in the spring ($\beta = 0.14$, $SE = 0.04$, $p = .001$). For language skills, there was no significant association between peers' EF in the fall and language in the spring. For pre-academic skills, peers' EF in the fall was a significant and positive predictor of children's early mathematics ($\beta = 0.06$, $SE = 0.03$, $p = .049$) and early literacy in the spring ($\beta = 0.09$, $SE = 0.04$, $p = .027$).

Interaction between classroom quality and peers' EF on gains in individual children' EF, pre-academic, and language skills

Interaction analyses examined whether the influence of peers' EF on individual children's EF, pre-academic, and language skills was moderated by the quality of their classroom experiences. An interaction term between peers' EF and classroom quality was added to the main effects model, including all the covariates and baseline skills for each outcome. The interaction between peers' EF and classroom quality was significant for literacy skills ($\beta = -0.07$, $SE = 0.03$, $p = .040$), indicating that when children are in low-quality classrooms (CLASS scores were 1 *SD* below the mean), having peers with higher EF was more strongly related to gains in literacy skills (see Fig. 1). Specifically, in low-quality classrooms, peer EF was significantly and positively related to literacy skills ($\beta = 0.15$, $SE = 0.05$, $p = .003$), but in high-quality classrooms (CLASS scores were 1 *SD* above the mean), peer EF was not significantly related to literacy skills ($\beta = 0.02$, $SE = 0.05$, $p = .686$). Contrary to our expectations, no other interaction effects were found between peers' EF in the fall and classroom quality (in the winter) in predicting gains in children's EF, language, and mathematics from fall to spring in the preschool (see Table 3).

Posthoc Analyses

Posthoc analyses were conducted to explore the extent to which subdomain of classroom quality was driving the effects of the broader interaction analyses. We tested the models separately for each subdomain of classroom quality and explored the potential underlying effects. Results showed that only the emotional support subdomain of classroom quality was statistically significant in the interaction. Specifically, the interaction between peers' EF and emotional support was significant for the literacy skills ($\beta = -0.07$, $SE = 0.03$, $p = .007$), meaning that when children experience low emotional support, having peers with stronger EF skills was more strongly related to gains in

literacy skills. In contrast, the interaction between peers' EF and instructional support ($\beta = -0.07$, $SE = 0.04$, $p = .051$) or classroom organization ($\beta = -0.015$, $SE = 0.03$, $p = .668$) was not statistically significant.

Discussion

This study examined the associations between peers' EF and gains in several aspects of school readiness skills during preschool. Consistent with theoretical (Bandura, 1997; Bronfenbrenner & Morris, 2007; Vygotsky, 1987) and empirical evidence (Choi et al., 2018; Justice et al., 2011; Mashburn et al., 2009; Montroy et al., 2016; Skibbe et al., 2012; Yeomans-Maldonado et al., 2019; Yudron et al., 2014) on the importance of the peers' skills for individual children's development, results from the current study broadly supported prior evidence by indicating that peers' EF was significantly associated with individual children's gains in EF, math, and literacy gains, but not language. Further, there was a significant interaction between peers' EF and the quality of the classroom environment for children's literacy gains, such that for children who are in low-quality classrooms, having peers with higher EF was more strongly related to gains in literacy scores. The interaction between peers' EF and classroom environment was not significant for EF, language, and math gains. A more nuanced understanding of the peer effects on children's development and the extent to which these associations interact with the classroom quality can help to better structure children's early learning environments by informing policy and practice.

Peers' EF and Children's EF, Pre-academic, and Language Skills

The current findings revealed differential associations between peers' EF and children's school readiness outcomes. As hypothesized, peers' EF was significantly related to individual children's EF gains across the preschool year. Specifically, children in classrooms where peers have higher EF at preschool entry demonstrated higher EF skills themselves at the end of preschool. Notably, these associations emerged in the current study even after controlling for various child-, peer-, and classroom-level covariates, including individual EF skills in the fall, age, income, sex, as well as their peers' age, income, and the quality of the classroom environment. Our findings contradict findings from a previous study documenting a non-significant link between peers' EF and individual children's EF (Weiland & Yoshikawa, 2014), yet support

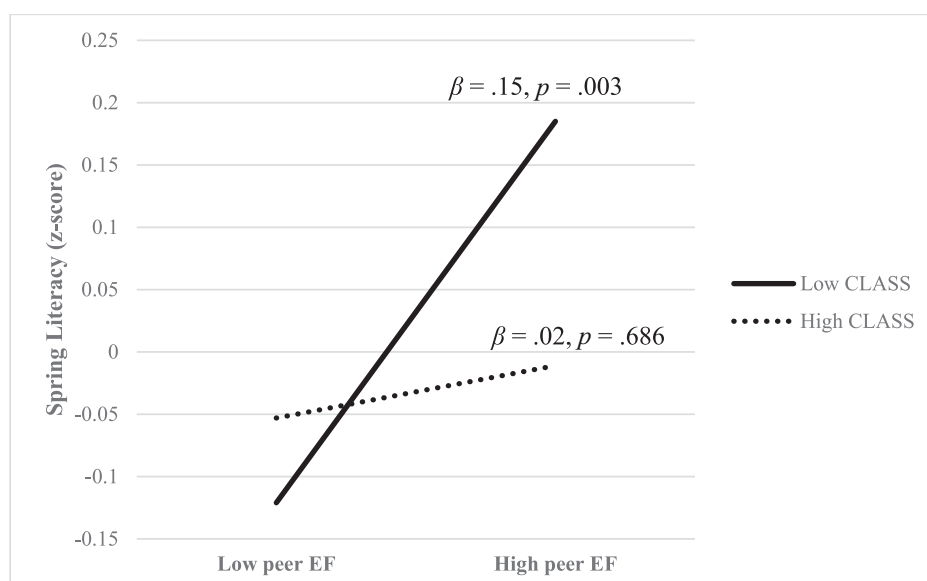


Fig. 1. Interaction between Classroom Quality and Peers' EF for Children's Gains in Literacy Skills
Note. EF = Executive function, CLASS = Classroom Assessment Scoring System. Low peer EF refers to when classroom-level EF was 1 *SD* below the mean and high peer EF refers to when classroom-level EF was 1 *SD* above the mean. Low CLASS refers to classrooms where CLASS scores were 1 *SD* below the mean and high CLASS refers to classrooms where CLASS scores were 1 *SD* above the mean.

recent findings with similar samples demonstrating significant links between peers' EF and individual children's EF in preschool (Choi et al., 2018; Montroy et al., 2016). One potential explanation for the discrepancy in prior literature may be that Weiland and Yoshikawa's (2014) study was in fact an evaluation of an intervention that aimed to improve children's EF, therefore they argued that their program may have led to improvements in children's EF skills overall for the entire classroom. Our sample demographics and study design were similar to the studies documenting significant links (Choi et al., 2018; Montroy et al., 2016), thus contribute to the understanding of peers' influence on children's EF by robustly testing the associations with a diverse sample.

In addition to links between peers' EF and children's EF, our findings demonstrated significant, albeit small, associations between peers' EF and gains in math skills over and above their individual EF and math skills in the fall and other covariates. Our finding is consistent with prior evidence documenting strong associations between EF and math skills (Clements et al., 2016; Cragg & Gilmore, 2014; Duncan et al., 2007; Schmitt et al., 2017), and extends previous literature by highlighting the importance of peers' EF for children's development of math skills, even though the effect size was small. The extent to which peers influence children's learning and development has long been argued (Henry & Rickman, 2007), yet the implications of peers' EF for individual children's development of math over the preschool year has not been previously studied. The findings imply that when children are in classrooms with higher EF skills at the beginning of the preschool, they demonstrate higher math skills at the end of the preschool, and this association is independent of peers' income, prior EF and math skills, or the quality of the classroom environment.

Similar to our findings related to math, our results also documented significant evidence suggesting that peers' EF was related to children's literacy gains, controlling for individual EF and literacy skills in the fall and other study covariates, including child- and peer-level covariates and classroom quality. The finding is in line with previous work that has identified significant, but small associations between peers' EF and individual literacy skills in preschool (Montroy et al., 2016). When children have peers with higher levels of EF in their classroom, their literacy skills over the preschool period may be supported irrespective of their own EF skills. In fact, our findings suggest that peers' EF was related to children's literacy gains, but their own EF skills in the fall were not. Although this is an unexpected finding, it is consistent with previous literature documenting a weaker association between EF and literacy skills compared to EF and math skills (Hassinger-Das et al., 2014; Schmitt et al., 2014).

With respect to our findings for language, we found no evidence that peer's EF was related to children's language gains over the preschool year. Children's individual EF and language skills in the fall were significant predictors of language gains. Despite the theoretical connections supporting an association between peers' EF and language, the literature was mixed, and our findings align with recent studies that documented no significant cross-domain associations between peers' EF and language gains (Choi et al., 2018; Weiland & Yoshikawa, 2014). The lack of effects of peer EF on gains in children's end-of-preschool-year language skills may be due to how children's language skills were measured. Both in our study as well as in the two studies (Choi et al., 2018; Weiland & Yoshikawa, 2014) that documented non-significant associations between the two, a receptive vocabulary measure was used to assess language skills. Although children's receptive and expressive language skills are strongly correlated (Gómez et al., 2019), it is argued that children's expressive language rather than receptive may be more influenced by their peers' EF (Weiland & Yoshikawa, 2014). Yet, Justice et al. (2011) found no significant main effects of peers' language skills (including receptive and expressive vocabulary) on children's language outcomes. Given these mixed findings and the importance of language development in early childhood, further research that examines the associations between peers' EF and receptive and expressive vocabulary skills is needed to enhance our understanding

of this association.

Interaction between Classroom Quality and Peers' EF on Children's EF, Pre-academic, and Language Skills

Another goal of the current study was to investigate whether the effects of peers' EF on children's development of EF, math, literacy, and language skills were moderated by classroom quality. Somewhat contrary to our expectations, only the association between peers' EF and literacy gains was moderated by classroom quality. However, for EF, language, and math skills, no significant interaction effects were found between peers' EF and classroom quality. Specifically, our findings demonstrated that the association between peer EF and their literacy gains across the preschool year was stronger for children who are in low-quality classrooms. In fact, posthoc analyses revealed that only the emotional support subdomain of classroom quality was statistically significant in the moderation analyses. It is argued that the emotional support subdomain of classroom quality is mostly related to children's social-emotional outcomes (Downer et al., 2010; Hamre et al., 2014), yet our findings demonstrated the comfort teachers provide in the classroom, including expressing warm and positive feelings and respect toward children and being sensitive and responsive to their needs, also help children's development of pre-academic skills, such as their literacy skills. The current study findings imply that having peers with high EF in low-quality classrooms, especially when teachers' emotional support was low, was more strongly related to their literacy gains across the preschool year than having peers with high EF in high-quality classrooms. Prior research has demonstrated that children's learning over the preschool year is related to classroom quality and children show the least amount of growth when they are in classrooms classified as low quality (Curby et al., 2009). Findings from the current study expand prior knowledge by demonstrating a synergistic association between peers' EF and classroom quality, indicating that in classrooms where there is low quality, especially emotional support domain of classroom quality, peer EF was more strongly related to children's development of literacy skills in preschool than where there is high quality.

Limitations and Future Directions

Our findings contribute to an emerging literature that considers the role of peers' EF in the peer effect literature. Our findings may have implications for children from families with low incomes and public preschool programs, yet the generalizability of the findings should be kept in mind. Further research is needed to replicate these findings with different classroom compositions. Although children's outcome measures and classroom quality relied on direct assessments and observations, our measure of language was limited to receptive vocabulary, and EF was limited to one assessment. Future research including expressive vocabulary as a measure of language and multiple EF assessments may shed light on potential differential associations. As in previous peer effect studies, our study considered peer effects as an average of a classroom experience, with a relatively small number of children (on average seven) and could not measure individual peer interactions. Peer groups may fluctuate in a preschool setting, and the frequency and duration of peer interactions (e.g., interactions between close friends) could not be considered in the current study. In addition, the information on the total number of children per classroom was not available to investigate. A nuanced measure of peer interactions as well as the total number of peers per classroom may provide more accurate estimates of peer effects on children's development.

Moreover, due to the correlational design of the study, no causal inferences can be made between peer effects and children's outcomes. Future research could investigate the causal associations between peers' EF and EF, pre-academic, and language skills by employing experimental designs (e.g., random assignment to peer play groups by skill level). It is also important to acknowledge that children were not

randomly selected to participate in the study. Future research may benefit from a random assignment to better generalize the study findings to the population. In addition, peer EF was assessed in the fall, and classroom quality was assessed in the winter of the preschool year. Even though we controlled for classroom quality when examining the associations between peer EF and individual children's outcomes, classroom quality in the winter may have been affected by peer EF in the fall. Future research should examine the potential bidirectionality between peer EF and classroom quality using longitudinal data with multiple assessments of classroom quality. Finally, previous research has documented that peer effects are strongest for children with low (Justice et al., 2011; Montroy et al., 2016) or high initial skills (Choi et al., 2018). Although our posthoc analyses provided null findings for the interaction between peers' initial skill level and individual children's EF in the fall for EF, pre-academic, and language skills in the spring, future research could contribute to these disparate findings.

Conclusion and Implications

Overall, peers with higher EF skills may create opportunities that children can focus, play collaboratively without disruption (e.g., taking the toy away when frustrated), and problem-solve when faced with challenges, which in turn may boost their own EF skills. EF processes are considered as prerequisites for learning academic skills, including math and literacy, and practicing inhibition, holding information in mind, and shifting attention during interactions with peers may subsequently boost their pre-academic skills. In addition, having peers with high EF skills in the classroom may assist individual children's learning pre-academic skills in various interactions including small and large group activities (Montroy et al., 2016). For instance, peers with high EF skills may be more likely to conform to classroom rules and expectations (e.g., waiting to take turn), allowing children to take advantage of learning opportunities related to classroom instructions such as explicit math or literacy instruction or to take initiative to interact with others to benefit from peer-to-peer or adult-to-child interactions. These early experiences may help children's transition and adaptation to formal school where there is a more structured environment and rules they are expected to follow. Furthermore, having peers with higher EF skills buffered the potential negative influences of low classroom quality on literacy outcomes, highlighting the important role of peers in the development of pre-academic skills.

These findings may have implications for intervention and practice in early childhood classrooms. Intervention efforts aiming to boost children's EF at the classroom level could benefit not only the targeted EF but the math and literacy skills of individual children as well as their peers. Further, our findings indicated that having peers with higher EF skills buffered the influences of being in a low-quality classroom for the development of literacy skills. These findings are especially relevant for early childhood programs that serve low-income children (e.g., Head Start). Given the homogeneous structure of these programs (Weiland & Yoshikawa, 2014) and the likelihood of starting with lower EF skills (Blair & Raver, 2015), targeted efforts for a heterogeneous preschool enrollment (e.g., mixed-income groups), intentional peer group structures including children with varying degree of EF skills, as well as interventions to boost EF skills and the classroom quality (especially emotional support) may benefit the development of EF and pre-academic skills for all children in a classroom.

Declarations of Competing Interest

None.

CRedit authorship contribution statement

Irem Korucu: Conceptualization, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing.

Tanya M. Paes: Writing – review & editing. **Lauren Costello:** Writing – review & editing. **Robert J. Duncan:** Funding acquisition, Formal analysis, Project administration, Writing – review & editing. **David J. Purpura:** Funding acquisition, Project administration, Writing – review & editing. **Sara A. Schmitt:** Conceptualization, Funding acquisition, Project administration, Resources, Writing – review & editing.

Data availability

The authors do not have permission to share data.

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